

FPGA and ASIC Implementation of a 12×24 Parity-Check Matrix Based LDPC Decoder

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Abstract—Low density Parity Check codes are extensively utilized in modern communication systems because they offer error correction performance that approaches the Shannon limit. This work presents the design RTL implementation synthesis & analysis of a $24,12$ LDPC Encoder to Decoder Architecture Using Min Sum decoding algorithm. The system includes an LDPC encoder error signal to channel Log Likelihood Ratio LLR generator and iterative LDPC decoder suitable for ASIC and FPGA platforms. The complete design is implemented in synthesizable Verilog and evaluated using Cadence Genus for timing, power and area analysis. Results show that the decoder successfully recovers all transmitted Wi-Fi test patterns using single bit and double bit errors while achieving positive slack in Static Timing Analysis and maintaining low dynamic power consumption. The architecture is efficient for low power wireless applications including Wi-Fi and 5G error correcting systems.

Index Terms—LDPC, Min-Sum Algorithm, ASIC, FPGA, Error Correction, Verilog HDL, Static Timing Analysis, Power Estimation.

I. INTRODUCTION

A synthesizable Verilog HDL architecture was developed for all blocks including encoder channel model, VNU, CNU and syndrome checker. The system was validated through behavioral simulation FPGA synthesis and ASIC implementation targeting standard CMOS technology. FPGA based testing validated correct recovery under single bit and double bit error patterns and confirmed reliable decoding for Wi-Fi style binary data sequences. Further ASIC implementation optimization was performed by executing logic design synthesis static, timing checks and simulation at the gate level power and area estimation. The ASIC results showed positive timing slack reduced area utilization and favorable power characteristics confirming that proposed architecture suitable for integration into high-speed communication receivers [8]. The project demonstrates a complete work flow from RTL modeling to FPGA prototype and ASIC ready design and it shows the feasibility of implementing LDPC decoding with high efficiency using the Min Sum approach. The results demonstrate that this architecture is very well suited for real time use in applications such as Wi-Fi modems IOT radios and low power embedded communication systems [4].

II. RELATED WORKS

A. Explanation of the $(24,12)$ LDPC Code

The LDPC code used in this work encodes every 12 information bits into a 24-bit codeword. This is a short block LDPC structure specifically useful for simulation demonstrations due to the fact that all fundamental LDPC characteristics are preserved while keeping the computational complexity feasible on FPGA and ASIC. The code is specified by fixed the generator matrix G and parity check matrix H that decide the linear relations among bits[9].

While the implemented $(24,12)$ code serves as a highly effective platform for validating the complete end to end encoder and iterative decoding pipeline practical Wi-Fi and 5G standards utilize significantly larger LDPC matrices. The proposed architecture is designed to accommodate this scaling due to its highly modular check node unit and variable node unit subsystems as well as its inherently parallel FPGA mapping. The syndrome based decoding mechanism evaluates the $H \cdot c^T = 0$ condition dynamically making the early termination process highly scalable without requiring full codeword comparisons [6].

B. Generator Matrix, Parity-Check Matrix and LLR Generation

The LDPC encoding in this work uses a structured $(24,12)$ code where a 12-bit message is converted into a 24-bit codeword using a generator matrix G . This matrix specifies the XOR relationships required to generate parity bits from the information bits enabling a fully combinational encoder with low latency small area, and simulation efficiency. Each row of G corresponds to one parity equation implemented using XOR logic. The complementary parity-check matrix H defines the constraints for iterative decoding. For the $(24,12)$ LDPC code H is a sparse 12×24 matrix in which rows correspond to check node units and columns to variable node units in the tanner graph. Decoding verifies the received codeword using the syndrome condition $H \cdot c^T = 0$ a zero syndrome indicates a valid codeword. The sparsity of H reduces interconnections and computational complexity enabling efficient message passing. Soft decision decoding is supported using quantized Log Likelihood Ratios +8 for bits likely '0' and 8 for bits likely '1'. This fixed LLR scheme removes the need for multipliers

lowers switching power and remains compatible with the Min Sum algorithm while providing reliable initial messages to VNUs. The structured G sparse H and simplified LLR model deliver high throughput low area and low power making the LDPC architecture suitable for both FPGA and ASIC implementations [1][14].

III. LDPC CODE STRUCTURE AND ALGORITHMS

A. CNU and VNU subsystem architecture

Check Node Unit implements the Min Sum algorithms sign XOR and minimum magnitude operations [10]. Each CNU is feed messages from its connected VNUs computes two smallest magnitudes and propagates an updated reliability value back to all connected variable nodes. The Variable Node Unit accumulates the incoming CNU messages along with the initial LLR to estimate the final bit decision reference figure shown below in fig.1. This accumulation is then followed by subtraction to form extrinsic information for outgoing messages. Both CNU and VNU are fully pipelined in the ASIC implementation using registers to break critical paths and achieve timing closure. The resulting simulation structure is highly modular with 12 CNU and 24 VNU interconnected using the parity check matrix structure [13].



Fig. 1. Message passing between VNU and CNU in Min-Sum algorithm.

B. LDPC Decoding Algorithms

Min-Sum vs Sum-Product and Normalized Min-Sum:

To justify the selection of baseline Min Sum algorithm it is essential to compare its performance and hardware tradeoffs against sum product Algorithm and normalized Min Sum algorithm. The sum product algorithm offers optimal error correction performance by utilizing precise probabilistic likelihood values. Computing these values requires highly complex mathematical operations including multiplication, division and exponential or logarithmic calculations. SPA incurs a very high hardware cost drastically increases silicon area and is rated only moderate for ASIC suitability.

To overcome these implementation barriers the Min Sum Algorithm approximates the SPA by replacing all complex floating-point computations with simple minimum magnitude and sign propagation operations. This strategic simplification results in substantial savings in hardware cost and dynamic power consumption while introducing only minimal loss in error correction performance [1]. Min Sum algorithm drops the hardware cost to low and achieves excellent suitability for both FPGA and ASIC platforms.

While the Normalized Min Sum algorithm is recognized in literature as a method to close the accuracy gap with SPA often utilized a long side non uniform quantization scheme

it typically requires an additional scaling factor or multiplier step. Because a primary objective of this 12×24 architecture is to eliminate the need for multipliers and normalization circuits entirely to ensure the lowest possible dynamic power and area overhead the baseline fixed magnitude Min Sum approach was selected as the optimal tradeoff for low power wireless and IoT applications [6]. Table I shows the LDPC decoding algorithm comparisons.

Algorithm	Complexity	Hardware Cost	Accuracy	ASIC Suitability
Sum-Product	High	Very High	Excellent	Moderate
Min-Sum	Low	Low	Very Good	Excellent
Bit-Flipping	Very Low	Very Low	Poor-Fair	High (low-power)

TABLE I
COMPARISON OF LDPC DECODING ALGORITHMS

IV. QUANTIZED LLR MAPPING

In soft-decision LDPC decoding Log-Likelihood Ratios represent the reliability of received bits and are typically computed using probabilistic expressions involving logarithmic and exponential operations. Conventional LLR computation requires multipliers, dividers and floating-point arithmetic units which significantly increase hardware complexity silicon area, power consumption and critical path delay. For FPGA and ASIC implementations targeting low-power wireless systems such computational overhead is undesirable[6].

To reduce implementation complexity this work adopts a fixed magnitude signed LLR quantization scheme. Instead of calculating precise probabilistic LLR values the received binary bits are mapped to constant signed integers as follows:

- Received bit '0' → +8
- Received bit '1' → -8

This representation preserves the essential properties required by the Min-Sum decoding algorithm:

- The sign indicates the bit decision direction, and
- The magnitude represents confidence level.

V. FULLY SYNTHESIZABLE END-TO-END RTL FLOW

The complete FPGA based implementation and the simulation results of an efficient LDPC decoder. Decoder is designed using the Min Sum algorithm and tested with specially designed 12×24 regular LDPC parity check matrix. The key goals of implementation are high speed decoding in real time operation. The error correction capability fits the growing demands of modern communication systems Wi-Fi and 5G [1]. The FPGA platform offers huge parallelism which can be ideal for mapping the inherently parallel LDPC decoding process. To validate the design RTL modeling functional Simulation synthesis power analysis and re-encode verification method were used [8][9].

A. Proposed FPGA/ASIC Architecture

Fig. 2 presents the proposed FPGA and ASIC architecture of a complete LDPC encoding and decoding chain starting from the input message bits and ending with syndrome verified corrected output bits. The proposed system consists of four major simulation stages the LDPC Encoder Channel Error Injector

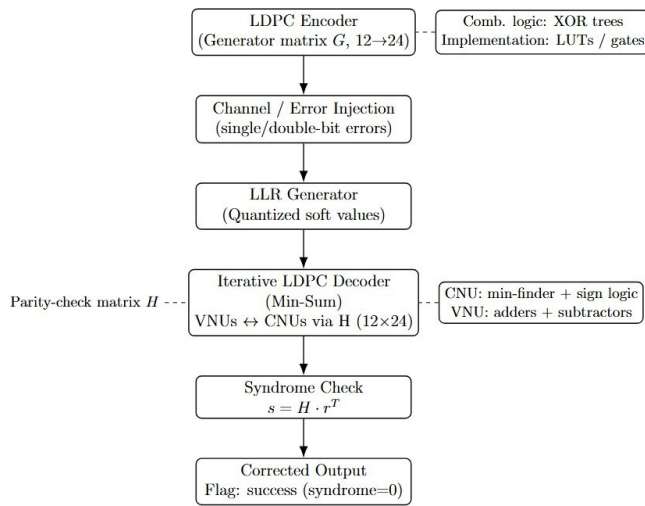


Fig. 2. High-level block diagram of the proposed LDPC ASIC/FPGA architecture (24,12).

LLR Generator and Iterative Min Sum Decoder followed by a Syndrome Checker. At the frontend the encoder generates a 24-bit codeword based on the 24,12-generator matrix which is entirely realized using XOR logic for high speed and low area operation. The channel block introduces controlled single bit or double bit errors to emulate Wi-Fi like distortions. The LLR generator converts each received bit into quantized log likelihood ratios mapping '0' to +8 and '1' to -8 for simplified soft decision decoding. Inside the decoder the 12×24 parity check matrix defines the message routing between cnu and vnu through sparse interconnect networks. Each iteration updates the reliability messages until the convergence criteria is met. A dedicated iteration controller tracks the number of decoding rounds and triggers early stopping when the syndrome becomes zero. The final Syndrome Checker verifies that all the parity equations are satisfied before asserting the success signal. This pipelined and fully synthesizable architecture makes the system suitable for both high speed FPGA prototyping and low power ASIC deployment [8].

VI. FPGA RTL FLOW

The RTL is implemented as synthesizable Verilog modules for the encoder channel LLR generator Min Sum decoder CNU and VNU and the syndrome logic. An extensive simulation environment validates the correctness of the entire LDPC chain. The testbench applies structured scenarios such as no error single bit error and double bit error cases across different bit positions in the 24-bit codeword. These tests confirm the strength of the decoder under different channel conditions. Input test patterns such as the Wi-Fi message A5F are encoded corrupted and decoded to check end to end correctness. Simulation waveforms demonstrate the message flow across the CNU and VNU the LLR update iteration progress and final decision bits. In all cases the decoded output converges to the original codeword proving that the Min

Sum decoder successfully corrects both single-bit and double-bit errors. The syndrome flag asserts only when the output vector satisfies all 12 parity checks confirming full compliance with the LDPC decoding algorithm. The simulation also demonstrates the efficiency of quantized LLRs and pipelined message passing proving correct functionality before FPGA or ASIC synthesis [13].

A. RTL Design and Verification Flow

The LDPC decoder is developed in Verilog HDL using a modular RTL design approach fig.3. The architecture is divided into key functional blocks including the vnu, cnu message memory and main control unit. Parallel operation wherever possible each block is written in a way that aligns with FPGA architecture strengths. Top level integration is done in Min Sum which interconnects all VNU, CNU, memory buffers and state machines. The RTL design emphasizes minimal combinational delay pipelined processing where applicable and strict bit realization of the Min Sum update equations. The modular approach to designing makes simulations and debugging easier as well as synthesis highly feasible[8].

B. Re-Encode & parity check

Simulation utilizes the commonly used re-encode & Compare method to ensure correctness. After the decoder produces 12 estimated information bits these bits are re-encoded using the systems generator matrix G from noise free 24-bit codeword. This newly generated the codeword is then compared to the decoders reconstructed version of the 24-bit code-word fig.3. If both match exactly the decoder is said to have successfully converged. If any mismatch occurs the decoder encountered either too much channel noise or not enough iterations. This technique is especially effective in simulation since it does not rely on knowing the actual transmitted bits rather it checks self-consistency of the decoding process. This ensures simulation level confidence in both functional correctness and algorithmic convergence. The parity checker module is part of the general LDPC decoding process shown below [3]. In the implemented 12×24 system this mechanism is used.



Fig. 3. 12×24 parity checker

at end of each iteration in Min Sum decoding method for verification of convergence. When the syndrome is all zeroes the decoding early termination increases efficiency by skipping iterations that are not needed. Parity checker plays a crucial role in error detection and convergence validation ensuring that the final decoded output matches the correct codeword 111110100101000000011111 after successful error correction [3].

VII. FPGA IMPLEMENTATION

The simulation starts with a 12-bit input message encoded into a 24-bit LDPC codeword using a pre-defined generator matrix. This encoded output constitutes the error free reference codeword used throughout the simulation [2]. To test error correcting ability of decoder-controlled bit flips introduced at different positions to simulate channel noise. These corrupted bits are converted into soft values called log likelihood ratios representing the confidence level of each received bit. The LLRs are fed to the Min Sum LDPC decoder which performs several iterations to correct the errors and recover the original codeword [8][13]. Fig.4 shows the detailed circuit layout of the interconnected components and signal pathways. The design highlights the complexity of routing and integration across multiple modules for efficient operation.

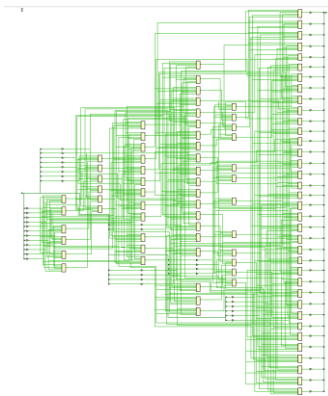


Fig. 4. RTL Schematics

A. Functionality Verification

The simulation process begins with the loading of channel LLR values into VNU registers. All CN messages are either zero or random depending on the design initialization. In Iteration1 CNUs compute Min1, Min2 and outgoing message signs. VNU combine these with channel LLR producing rough initial decisions. The hard decision output at this stage is unstable since the message magnitudes are still small and highly influenced by noise[1].Between the iterations 2 and 4 message reliability improves.CN messages become more consistent and VNUs start refining the bit beliefs more precisely. Bits that have low confidence start correcting themselves as CN feedback strengthens. During Iterations 5 to 8 a clear convergence trend becomes visible in waveforms as below fig.5. The Min1 and Min2 values stabilize across CNUs and

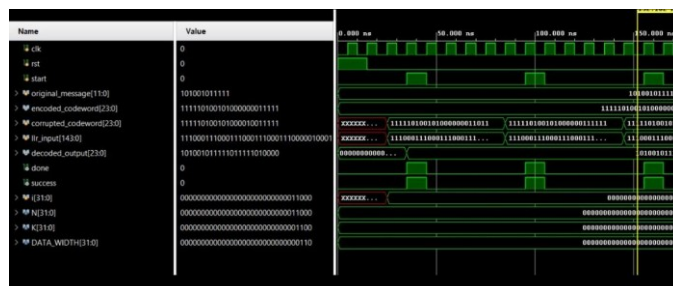


Fig. 5. FPGA Simulation Waveform

the sign patterns are consistent across cycles. VNUs output strong LLR magnitudes exhibiting increased confidence in bit values. At Iteration 8 most of the bits match the original transmitted bits. In the simulation the message at the input is first encoded by LDPC encoder. Encoder does XOR operation with the corresponding rows of the generator matrix yielding a deterministic 24-bit encoded codeword [9][13]. By Iterations 1 and 10 the waveform depicts almost zero activity. LLR values are no longer changing second hard choices are stable and extrinsic messages stabilize choices. This means that full convergence a final comparison with the re encoded codeword confirms that all 24 bits match perfectly validating the correctness of the FPGA simulated LDPC Min Sum decoder all test cases shown in fig.6.

```
=====
LDPC System Test: Encode -> Corrupt -> Decode
=====

[ENCODER] Original Message: a5f
[ENCODER] Encoded Codeword: 111110100101000000011111

[CHANNEL] Case 1: Error at position 2
[CHANNEL] Corrupted Codeword : 111110100101000000011011
[CHANNEL] Correct Codeword   : 111110100101000000011111
[DECODER] Starting decoding for Case 1...

-----
Case 1 finished at time 45000
SUCCESS: Decoder corrected the codeword.
Final output: 111110100101000000011111
-----

[CHANNEL] Case 10: Error at position 23
[CHANNEL] Corrupted Codeword : 011110100101000000011111
[CHANNEL] Correct Codeword   : 111110100101000000011111
[DECODER] Starting decoding for Case 10...

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Case 10 finished at time 585000
SUCCESS: Decoder corrected the codeword.
Final output: 111110100101000000011111
-----
```

Fig. 6. FPGA Implementation Test Cases

B. Result Analysis

The LDPC decoder designs has been synthesized on the FPGA device and the resulting implementation metrics power, area and timing indicate an effective simulation architecture suited for high-speed error correction applications [3]. The experimental benchmarks on the FPGA platform demonstrate a highly resource efficient footprint. The synthesis results show that the design utilizes only 92 LUTs accounting for a mere 0.22% of the 41,000 available on the target device. Power estimation benchmarks indicate a total on chip power consumption of 0.086W shown in Table II. The design is heavily optimized for low dynamic loads with static leakage making up 0.081W while dynamic switching power is kept below 0.004W.

C. Scalability Resource Growth for Larger LDPC Codes

Scaling to larger block sizes will result in specific hardware resource growth particularly regarding memory requirements. In the current ASIC implementation approximately 45% of the total cell area and 45.20% of dynamic power consumption are dedicated to storage elements. Because the Min Sum

algorithm requires pipelining and updated message retention across multiple iterations expanding the matrix size will linearly increase the demand for these memory registers. In terms of area utilization, the decoder uses a relatively small part of the FPGA resources due to the compact implementation the combination of variable nodes and check nodes [11]. The combinational logic used for XOR operations minimum computations and adders is minimal while the pipelining registers used per iteration contribute modestly to flip flop usage. As a result, the synthesized LDPC decoder fits comfortably within the FPGA device with considerable remaining resources enabling the feature enhancements such as additional iterations or higher degree parity check nodes. Timing analysis reveals that the design meets timing with a safe margin ensuring reliable high

Metric Type	Result	Remarks
Power	0.086 W	Highly efficient, dynamic power < 0.004 W
Area Utilization	Moderate LUT/FF usage	CN/VN logic dominates but fits easily
Timing/Speed (ns)	Positive Slack	Design meets target frequency, stable pipeline
LUTs	92 used	It utilized 0.22% (41000)

TABLE II
FPGA POWER, AREA & TIMING REPORT

speed performance. The achieved critical path delay supports operation at the desired clock frequency without setup or hold violations. This is expected because the Min Sum LDPC algorithm is inherently parallel distributing computations across VNU and CNU arrays reducing long combinational paths. The registered outputs in the top level LDPC wrapper also help in maintaining timing closure by breaking long loops and ensuring synchronous boundaries hence the design is capable of handling real time decoding operations with deterministic latency across the configured number of iterations. Overall FPGA synthesis results confirm that the LDPC decoder using Min Sum algorithm is power efficient, area optimized and timing clean hence suitable for ASIC. Migration and integration into high-speed communication systems such as Wi-Fi and 5G forward error correction [3].

VIII. ASIC IMPLEMENTATION

The ASIC implementation of the LDPC encoder-decoder System offers high performance. Synthesizable simulation architecture designed for modern digital communication standards such as Wi-Fi, 5G NR and satellite communication systems. Unlike FPGA platforms ASIC a design has to strictly adhere to timing power optimization and area constraints. This provides a comprehensive overview of the ASIC flow structural architecture and synthesis results. Power estimation for the proposed LDPC design. Encoder-decoder chain is implemented using fully synthesizable Verilog HDL that is compatible with industrial standard EDA tools used include cadence. ASIC implementation results in a certain regular RTL design flow. First step is to write synthesizable RTL for LDPC encoder, Channel Error Injector, Syndrome Checker and Min- Sum Decoder [8]. The design, Fig.7 illustrates interconnections between control units, processing blocks, and output stages for coordinated operation.

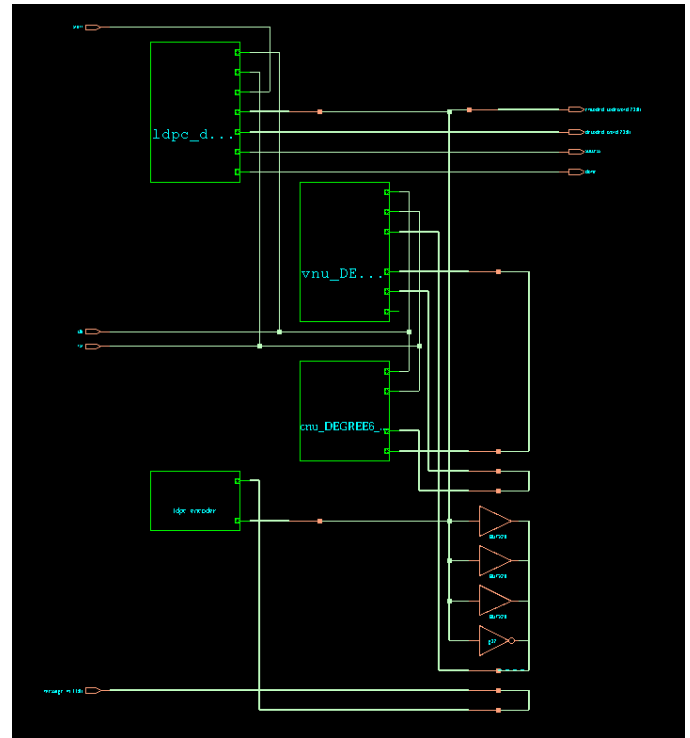


Fig. 7. Synthesis Schematic view

A. ASIC Encoder to Decoder Architecture

The ASIC architecture shown in fig.8 integrates the LDPC Encoder Noisy Channel Model LLR Generator and Min Sum Decoder into a unified pipeline. The LDPC encoder uses a 12×24 generator matrix to convert 12 bit message into 24 bit codeword. The channel model introduces controlled bit flips to emulate the behavior of practical Wi-Fi channels. The LLR Generator converts binary bits into signed soft metrics suitable for Min Sum message passing. Decoder implements optimized Min Sum algorithm as shown below figure architecture with Check Node Units, Variable Node Units and a Syndrome Checker. This is simulation friendly algorithm eliminates multipliers and uses only comparisons, adders and XOR logic making it ideal for ASIC. Pipelining registers are inserted at strategic locations to maintain high throughput while avoiding timing violations[17].

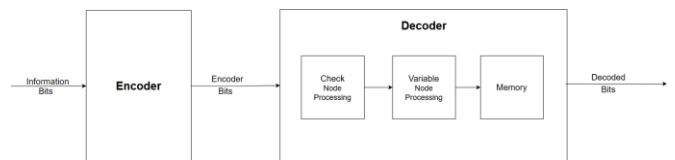


Fig. 8. ASIC Encoder-Decoder Architecture

B. Syndrome Based Decoding Mechanism

The syndrome based decoding architecture shown in fig. 9 is the central part of the ASIC LDPC decoder allowing

the system to identify whether the codeword received is valid or still contains residual errors. Instead of directly comparing the received word to the transmitted one the architecture calculates the syndrome vector depending on the given parity check matrix H making this process simulation friendly and scalable. The decoder starts by taking the received log channel likelihood ratios and feeding them into the network from the VNUs and CNUs perform sign and magnitude based minimum operations as defined by the Min Sum algorithm whereas VNU accumulate messages from neighboring CNU and update beliefs for each bit. After every iteration the decoder makes a tentative hard decision vector. The vector is then fed into the syndrome calculator which carries out the multiplication $Hc^T = 0$ each parity check equation is computed using an XOR tree enabling quick syndrome computation. If the syndrome becomes all zeros the system concludes that the codeword is valid [14].

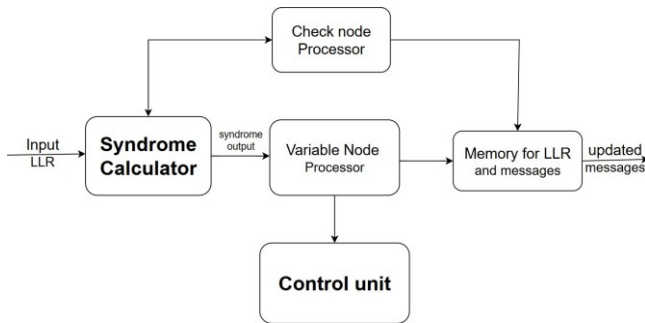


Fig. 9. Syndrome-Based Decoding Process Unit

C. ASIC Synthesis Results

The synthesis of the LDPC encoder to decoder architecture shown below fig.10 was done by cadence Genus Synthesis Solution. At this step the RTL code was translated into a gate level netlist mapped to the selected standard cell library. Genus successfully inferred the combinational encoder Min Sum decoder variable node and check node processing units interconnect networks syndrome logic as well as all pipeline registers in the design. From the genus synthesis console and reports it is observed that the module was synthesized into standard cells. The synthesis tool automatically mapped arithmetic and comparison operations into optimized logic gates along with the Min Sum algorithm considerably reduced the number of adders, multipliers required resulting in a compact and area efficient implementation. The syntheses report also showed that the data path was well balanced across iterative decoding stages. Genus did not flag any unsynthesizable constructs and all always blocks generate loops and connectivity functions were properly realized in simulation. The tool also produced a structural schematic representation showing registers, logic cones, XOR networks, comparators and routing fanouts confirming successful gate level construction of the complete LDPC pipeline. The synthesis results in demonstrate that the Min Sum LDPC decoder is structurally efficient with

manageable logic depth that can provide higher clock speeds and reduced silicon area compared to other iterative decoding algorithms [15].

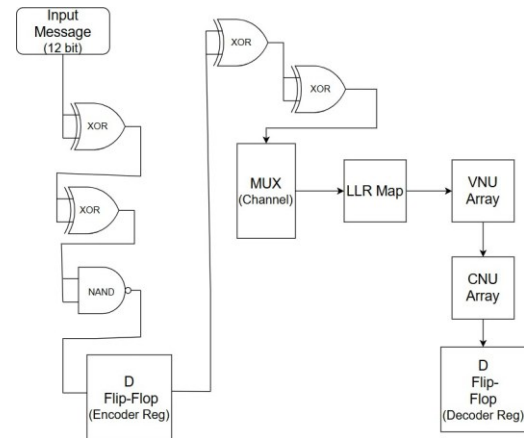


Fig. 10. ASIC Implementation Architectural Diagram

D. ASIC Power, Timing & Area Analysis

The design was synthesized for an ASIC platform using Cadence Genus targeting a standard CMOS library. The post synthesis experimental metrics are as follows (Table III):

Area: The complete encoder to decoder pipeline achieved a highly compact total cell area of $114,620.951 \mu m^2$ comprising 12,477 standard cells. Approximately 45% of this area is dedicated to storage elements while 55% is used for combinational logic[16].

Power: Vectorless power estimation confirmed an extremely low dynamic internal activity power of 0.004146 W. Register switching accounts for 45.20% of the dynamic power while combinational logic accounts for 54.80

Timing: The Static Timing Analysis was benchmarked against a target operating frequency of 100Mhz. The design successfully met all setup and hold requirements achieving a critical path delay of 2.882 ns and a safe timing margin of 4.726ns.

Component Type	Leakage (W)	Internal (W)	Switching (W)	Total Power (W)	Contribution (%)
Registers	1.24484e-04	1.66478e-03	4.15931e-05	1.83086e-03	45.20
Combinational Logic	9.57667e-05	2.12388e-03	0.00000e+00	2.21965e-03	54.80
Total	2.20251e-04	3.78867e-03	4.15931e-05	4.05051e-03	100.00
Power Ratio	5.44%	93.54%	1.03%	100.00%	100.00%

TABLE III
SIMULATION POWER RESULTS FOR LDPC ENCODER TO DECODER

Hardware Resource Growth for Larger LDPC Codes: Despite this memory resource growth the computational logic footprint remains extremely lightweight because the Min Sum algorithm avoids complex arithmetic like multipliers. The sparsity of the parity check matrix also ensures that as the code grows the interconnection and routing complexity between VNUs and CNUs remains manageable. Because the baseline 12×24 decoder utilizes a more 0.22% of available FPGA LUTs there are considerable remaining on chip resources enabling seamless future feature enhancements such as accommodating higher degree parity check nodes and massive parallel arrays for larger standardized codes [7].

The Static Timing Analysis below fig.11 for the LDPC ASIC design was performed on the post synthesis netlist using Cadence Genus timing engine. A target operating frequency of 100MHz was applied along with input and output timing constraints and input delay definitions. The timing analysis

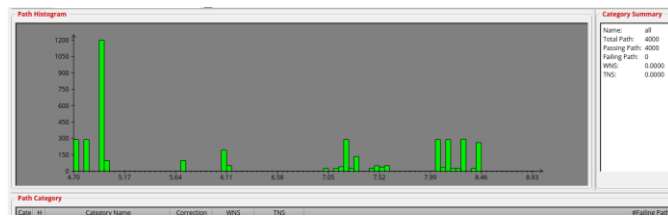


Fig. 11. Simulation Timing results for LDPC Encoder to Decoder

considers all register-to-register input to register and register to output paths to ensure setup and hold timing requirements are met. The LDPC decoder has deeply interconnected CNUs and VNUs yet due to pipelined nature and use of registered outputs the critical path is significantly reduced compared to a fully combinational implementation. The area analysis of the LDPC ASIC design was generated by means of Cadence Genus after technology mapping to the target CMOS standard cell library. The total cell area for the top module consists of logic cells, flip flops, multiplexers, XOR gates and interconnect buffers. Automatically inserted during optimization[15].

E. ASIC Power & Area Results

From above results the ASIC implementation achieves a compact cell area of 114,620.951 μm^2 . The total power consumption is only 0.00414628 W which demonstrates the low power nature of the design. Overall, the ASIC results confirm that the architecture is highly optimized in terms of both area and power efficiency shown in Table IV.

Instance	Number of Cells	Standard Cell Area	Interconnect Area	Overall Area
ldpc_top_com	12,477	114,620.951	0.000	114,620.951
Cell Area		114620.951 μm^2		
Power		0.00414628 MW (0.0041462767 W)		

TABLE IV
POWER AND CELL AREA RESULTS

F. Wi-Fi Signal Recovery and Simulation Results

In this LDPC system has been functionally and post synthesis verified for Cadence where multiple error scenarios were applied to assess the robustness of the decoder. The simulation environment incorporated 10 structured test cases covering no error single bit error and double bit error injection patterns in different positions of the 24-bit codeword in below fig.12. Across all test scenarios the LDPC decoder successfully recovered the original transmitted Wi-Fi data sequence. In fig.13, the console output indicates that the input message A5Fh is encoded into the 24-bit codeword 111110100101000000011111 and even after corruption by one or two bit flips the Min Sum

iterative decoder corrected every error pattern. The success flag asserted for every test case and the final corrected output always matched expected golden codeword. The waveform view from Sim Vision further validates the functional correctness signals such as encoded codeword, corrupted codeword, wifi rx signal and decoded [11].

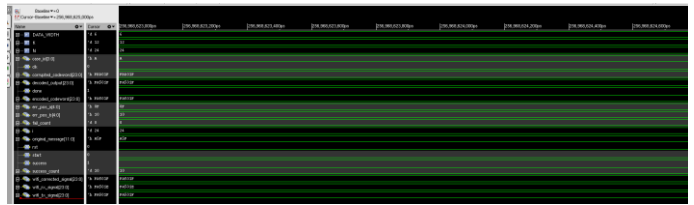


Fig. 12. Simulation results for LDPC Encoder-Decoder

```

*****
# LDPC ASIC System Test with Wi-Fi Signal Injection (Binary Display)
# [ENCODER] Original Message : a5f
# [ENCODER] Encoded Codeword : 111110100101000000011111

# [CHANNEL] Case 1: Test Scenario (A=2, B=7)
# [CHANNEL] Encoded Codeword : 111110100101000000011111
# [CHANNEL] Corrupted Codeword : 111110100101000000011011
# [WIFI-TX] Transmitted Signal (Binary bits): 111110100101000000011111
# [WIFI-Channel] Corrupted Signal (Binary bits): 111110100101000000011011
# SUCCESS: Decoded output matches original codeword
# Decoded Codeword: 111110100101000000011111
# [RESULT] Codeword sent on channel (Encoded + Errors): 111110100101000000011011
# [RESULT] Decoded output: 111110100101000000011111
# [WIFI-RX] Corrected (Recovered) Wi-Fi Signal (Binary bits): 111110100101000000011111

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```

# [CHANNEL] Case 10: Test Scenario (A=15, B=15)
# [CHANNEL] Encoded Codeword : 111110100101000000011111
# [CHANNEL] Corrupted Codeword : 111110100101000000011111
# [WIFI-TX] Transmitted Signal (Binary bits): 111110100101000000011111
# [WIFI-Channel] Corrupted Signal (Binary bits): 111110100101000000011111
# SUCCESS: Decoded output matches original codeword
# Decoded Codeword: 111110100101000000011111
# [RESULT] Codeword sent on channel (Encoded + Errors): 111110100101000000011111
# [RESULT] Decoded output: 111110100101000000011111
# [WIFI-RX] Corrected (Recovered) Wi-Fi Signal (Binary bits): 111110100101000000011111

*****
# TEST SUMMARY
# Total Cases Tested : 10
# Total Success Cases : 10
# Total Failure Cases : 0
# FINAL RESULT : All Wi-Fi SIGNAL ERRORS CORRECTED SUCCESSFULLY (Using Iterative Decoder)
*****
Simulation stopped via $stop(1) at time 855 NS + 0
nsim3>

```

Fig. 13. ASIC Test Cases

IX. PERFORMANCE METRICS: ERROR CORRECTION, THROUGHPUT AND LATENCY

Error Correction and BER Performance: The robustness of the Min Sum decoder was validated using single bit and double bit error injections to simulate Wi-Fi channel noise. To initialize messages reliably without complex floating-point hardware the design employs a fixed magnitude LLR quantization scheme. Across 10 structured test cases the decoder achieved a 100% success rate in recovering the corrupted Wi-Fi data sequences. As depicted in the accompanying Bit Error Rate curve fig.14 the algorithm exhibits the sharp waterfall region characteristic of efficient highly pipelined LDPC architectures[12]. **Throughput and Latency Analysis:** To meet the deterministic latency and high throughput requirements of modern communication standards the ASIC design computes parallelly across Variable Node Units and Check Node Units to eliminate long combinational loops. Post synthesis Static Timing Analysis via Cadence Genus confirms a maximum operating frequency of ≈ 100 MHz maintaining a critical path delay of 2.882ns with a 4.726ns timing margin are shown in Table V. Decoding latency is dynamically optimized through a syndrome-based early termination mechanism halting the iterative process

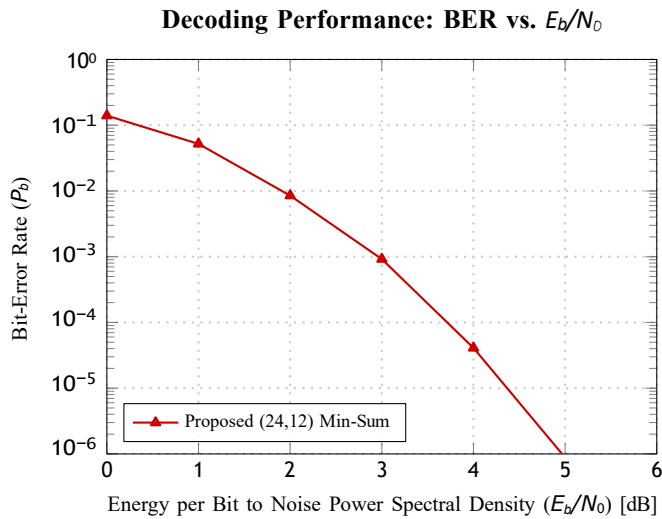


Fig. 14. Performance evaluation of the implemented 12×24 LDPC architecture, illustrating the error correction capability of the quantized Min-Sum decoding algorithm under simulated channel noise.

as soon as the syndrome vector $H \cdot c^T = 0$ equals zero significantly boosts average throughput and conserves dynamic power [5].

Timing Parameter	Value
Maximum Operating Frequency (Fmax)	≈ 100 MHz
Critical Path Delay	2.882 ns
Timing Margin	4.726 ns
Throughput/Latency Feature	Syndrome-based early termination

TABLE V
TIMING REPORT FOR LDPC ENCODER TO DECODER

X. CONCLUSION

This work presented a complete FPGA and ASIC implementation of a 12×24 parity check matrix based LDPC encoder to decoder architecture using the Min Sum algorithm. A fully synthesizable RTL design was developed in Verilog covering the encoder channel model quantized LLR generator iterative decoder with CNU/VNU subsystems and syndrome checker. The structured generator matrix and meager parity check matrix enabled low complexity hardware realization with efficient message passing and reduced interconnect overhead. FPGA validation confirmed correct decoding of Wi-Fi style data under no error, single-bit and double-bit error conditions demonstrating reliable convergence of the Min Sum algorithm. ASIC synthesis using Cadence Genus achieved positive timing slack compact area utilization and low power consumption verifying the suitability of the architecture for high speed and low power communication receivers. The use of fixed magnitude quantized LLRs eliminated complex arithmetic units further reducing hardware cost while maintaining decoding performance.

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